Lab 3: Shortest Paths in Graphs

Write a **Java program** that implements **Dijkstra's algorithm** to find shortest paths to all vertices in a directed **weighted** graph from a given starting vertex.

As mentioned in the lectures, choosing an implementation for the data structure representing a graph is highly dependent on the sparsity of the graph. For dense graphs, an adjacency matrix representation is preferable, but for sparse graphs, an adjacency list representation may be more efficient. You are asked to implement either the adjacency list or adjacency matrix representation for graphs in your program. If you choose to extend your implementation from Lab 2 then it will need account for **weighted edges**.

Your program should (a) read in data defining a graph from a **GraphML** file, (b) **build** an **adjacency list** OR **adjacency matrix** representation of the **weighted** graph and (c) use the graph representation to **find shortest paths** from a specified vertex in the graph to all other vertices, using **Dijkstra's algorithm***, as specified in the lectures and tutorials.

Getting Started

The starter code zip file contains sample code for parsing graphml data files, a `Graph` interface, which you are welcome to modify as you please, a driver class called `GraphShortestPathDriver` that can parse standard input, and a `GraphBuilder` class that can parse the data format described below. These classes should allow you to focus on the implementation of the Dijkstra's shortest path algorithm.

Data Format

There are a number of example graphml files in the repository provided as example input and output files. There files are located in the `data/graphs/` directory. For example:

- `data/graphs/graphSpecExample.graphml`
- `data/graphs/graphTutorialExample.graphml`
- `data/graphs/graph10.graphml`

Feel free to test your program with your own files as well. The data format uses the **GraphML**, which is described in detail here: http://graphml.graphdrawing.org/primer/graphml-primer.html

You could capture the output of your program and then use a diff program to compare to the expected output. This might be quicker than FLO Self-Review Quizzes (see below). IntelliJ IDEA has a nifty file comparison tool: https://www.jetbrains.com/help/idea/comparing-files-and-folders.html

And plenty of online diff tools can be found here, for example: http://www.mergely.com/editor

The **JDOM** library is an easy to use XML parsing library for Java. This library is included in the repository (see the lib/ directory). The binary download is available here: http://www.jdom.org/downloads/index.html

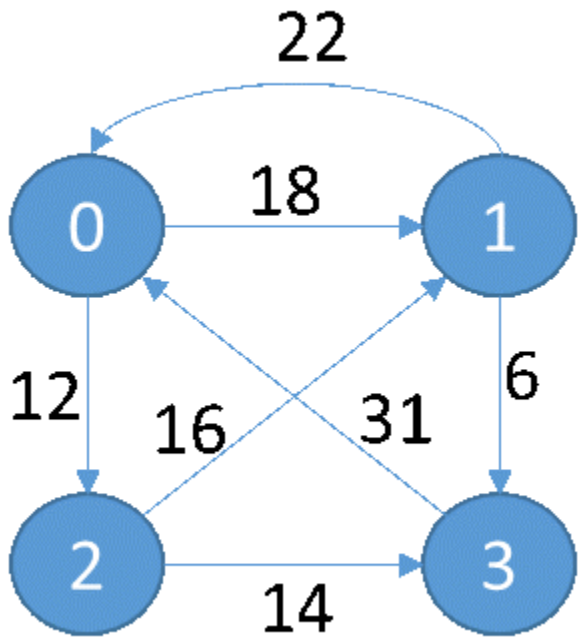
The XML file contains a list of vertices/nodes (the `<node>` element) and then a list of edges (the `<edge>` element). The weight for an edge is store as a child of the edge as `<data>` element with a `key` attribute of `"d1"` and the value of the data element is the weight. NOTE: All graphs start with a vertex with the "id" of "0" and increment by 1. This could be handy if you were using arrays or lists to represent key data structures in the graph shortest path algorithm...

For example, the contents of `graphSpecExample.graphml` is:

```
<?xml version="1.0" encoding="UTF-8"?>
<graphml xmlns="http://graphml.graphdrawing.org/xmlns">
  <key id="d1" for="edge" attr.name="weight" attr.type="int"/>
  <graph id="G" edgedefault="directed">
    <node id="0"/>
    <node id="1"/>
    <node id="2"/>
    <node id="3"/>
    <edge id="0" source="0" target="1">
      <data key="d1">18</data>
    </edge>
    <edge id="1" source="0" target="2">
      <data key="d1">12</data>
    </edge>
    <edge id="2" source="1" target="0">
      <data key="d1">22</data>
    </edge>
    <edge id="3" source="1" target="3">
      <data key="d1">6</data>
    </edge>
    <edge id="4" source="2" target="1">
      <data key="d1">16</data>
    </edge>
    <edge id="5" source="2" target="3">
      <data key="d1">14</data>
    </edge>
    <edge id="6" source="3" target="0">
      <data key="d1">31</data>
    </edge>
  </graph>
</graphml>
```

This means that there are 4 vertices, numbered 0 to 3. Vertex 1 has an edge to vertex 2, with weight 18, and an edge to vertex 3 with weight 12. Vertex 2 has an edge back to vertex 1, but its weight is 22 (because this is a directed graph, if there is an edge from A to B, it does not mean that there is an edge from B to A, or if there is, the weights do not need to be the same).

A visual representation of this graph is shown below:



Expected Output

If coded correctly, a call to find shortest paths from vertex 0 from the Graph above should produce the following output:

```
shortest path to 0: 0: cost = 0
shortest path to 1: 0 1: cost = 18
shortest path to 2: 0 2: cost = 12
shortest path to 3: 0 1 3: cost = 24
```

This lists each Vertex (in ascending lexicographical order) and for each Vertex, the path from the given vertex to that vertex and the associated cost with that traversal.

The graph in `graph10.graphml` has size of 10 vertices. If coded correctly, a call to find shortest paths from vertex 0 should produce the following output:

```
shortest path to 0: 0: cost = 0
shortest path to 1: 0 1: cost = 6
shortest path to 2: 0 6 8 9 3 7 4 2: cost = 35
shortest path to 3: 0 6 8 9 3: cost = 21
shortest path to 4: 0 6 8 9 3 7 4: cost = 28
shortest path to 5: 0 6 8 9 3 7 4 2 5: cost = 37
```

```
shortest path to 6: 0 6: cost = 6
shortest path to 7: 0 6 8 9 3 7: cost = 25
shortest path to 8: 0 6 8: cost = 11
shortest path to 9: 0 6 8 9: cost = 15
```

Self-Review Quiz

FLO will contain self-review quizzes to help test your assignment. The program must accept one line of input on the standard input that contains the name of the file to load in graphml format, a space, and then the index of the starting vertex (starting from 0). This is what the classes `GraphShortestPathDriver` and `GraphBuilder` are design to do. The output should be as above, but it should also include vertices that are unreachable from the starting vertex with the output of `NO PATH`, For example (user input indicated with `<-- user input`):

```
data/graphs/random_v10_e10_w50.graphml 0     <-- user input
shortest path to 0: 0: cost = 0
shortest path to 1: NO PATH
shortest path to 2: 0 2: cost = 49
shortest path to 3: 0 5 7 3: cost = 75
shortest path to 4: 0 2 4: cost = 93
shortest path to 5: 0 5: cost = 25
shortest path to 6: 0 5 6: cost = 29
shortest path to 7: 0 5 7: cost = 50
shortest path to 8: NO PATH
shortest path to 9: NO PATH
```

Submission of the Assignment

Submission will be via FLO. You should submit all source code files that make up your solution.

Use of Generative AI

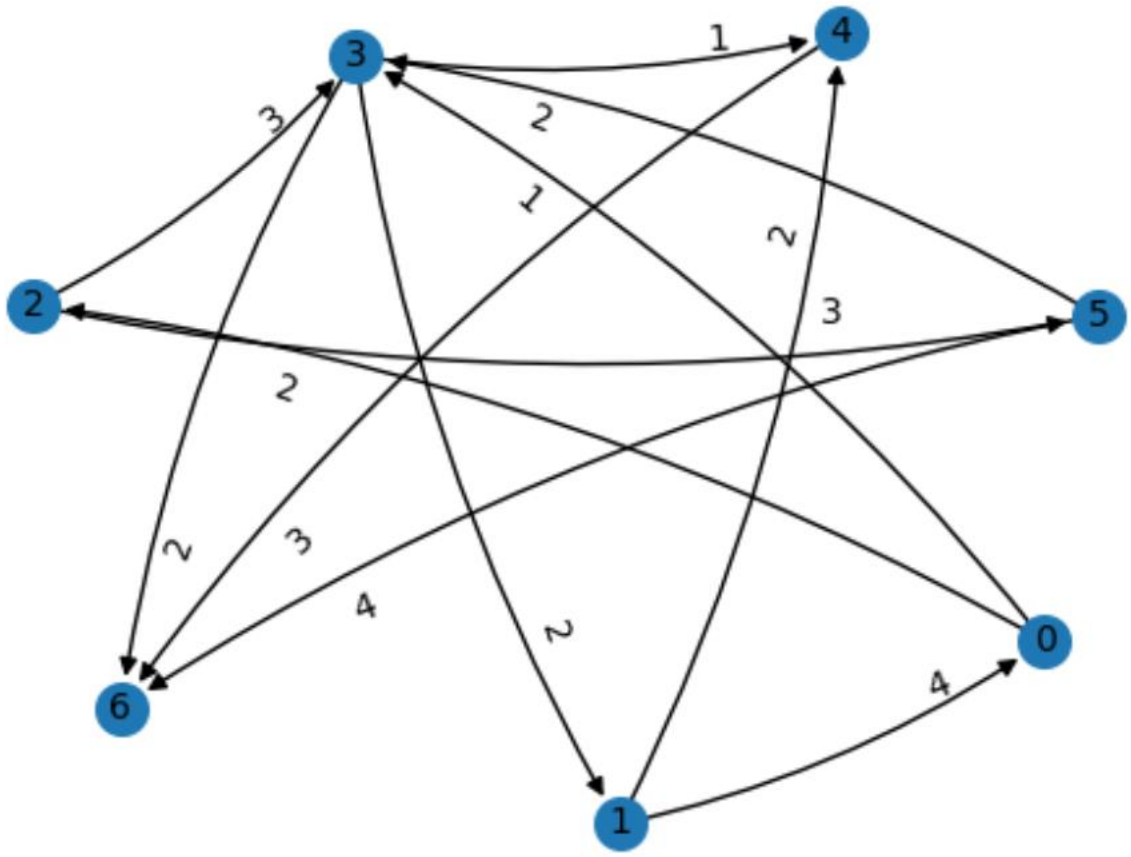
Note that AI **cannot** be used to write the code.
Also note that you should adhere to the normal university guidelines regarding the use of generative AI.

Additional Graph Visualisations

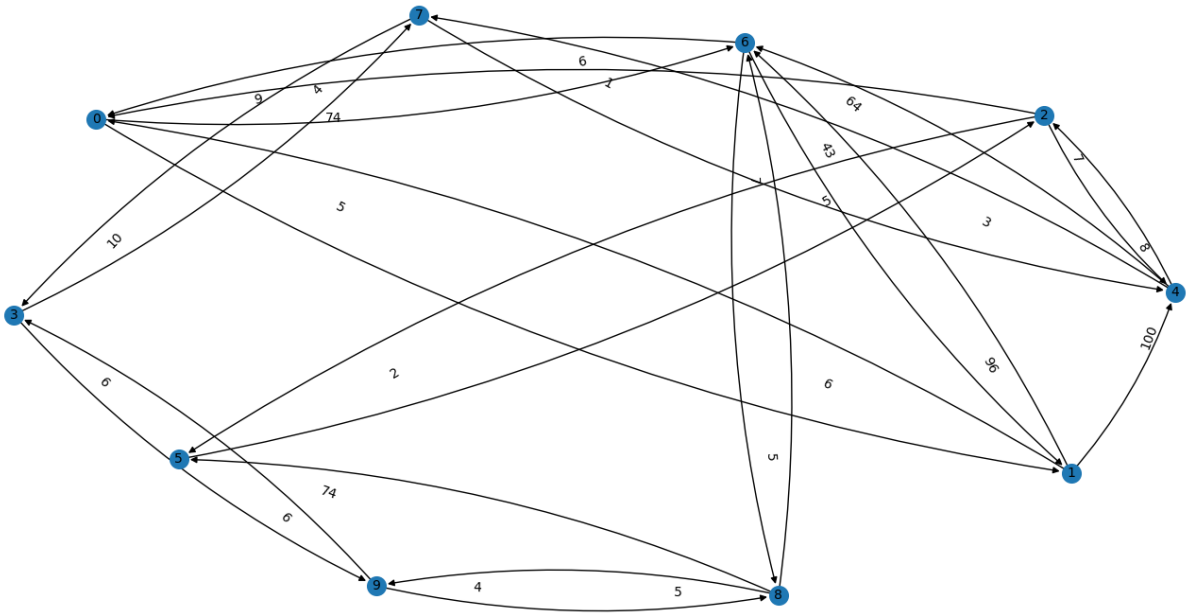
Below are some graph visualisations for some of the smaller example graphs in the data directory. The visualisations were created using Python and the `pygraphml` and `networkx` modules. The graphs are automatically laid out, so YMMV in the usefulness! If you are really keen you can check this out and a shortest path solver that creates a pptx of the steps out on Trent Lewis's Google Colab: https://colab.research.google.com/drive/13RGOJ1W3ACH-9zPKjH87Esv3YmD_Dc-B?usp=sharing. You will find the required `graph_template.pptx` on FLO.

Note, the code on the link is hastily hacked together and the graph data structure is an *edge list* which is an *inefficient* data structure for the shortest path algorithm. You should be using an adjacency list or adjacency matrix.

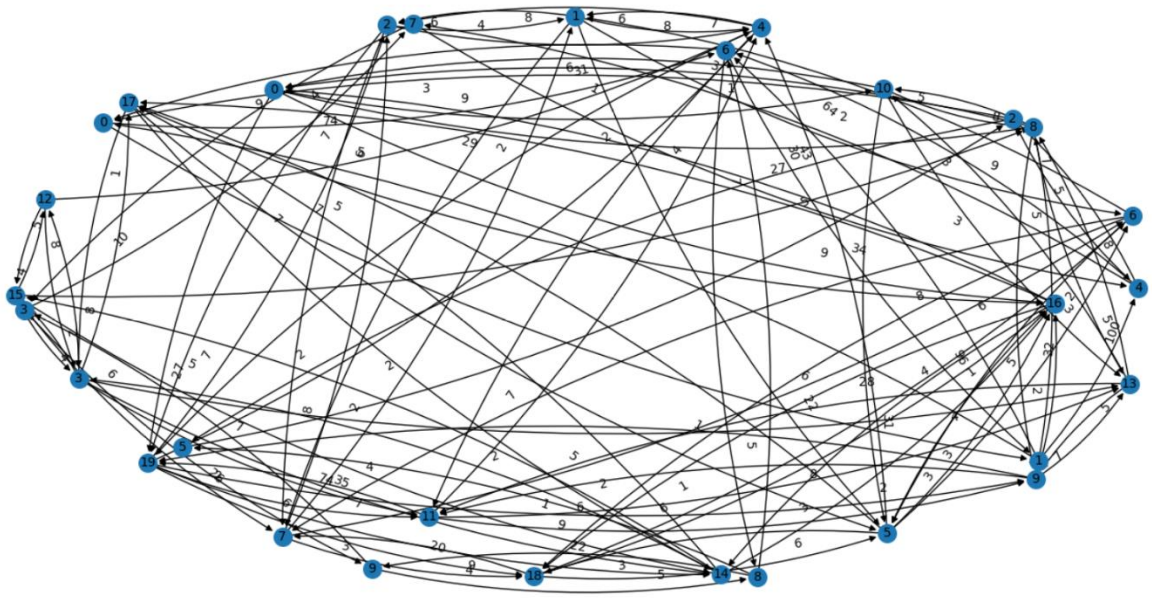
graphTutorialExample.graphml



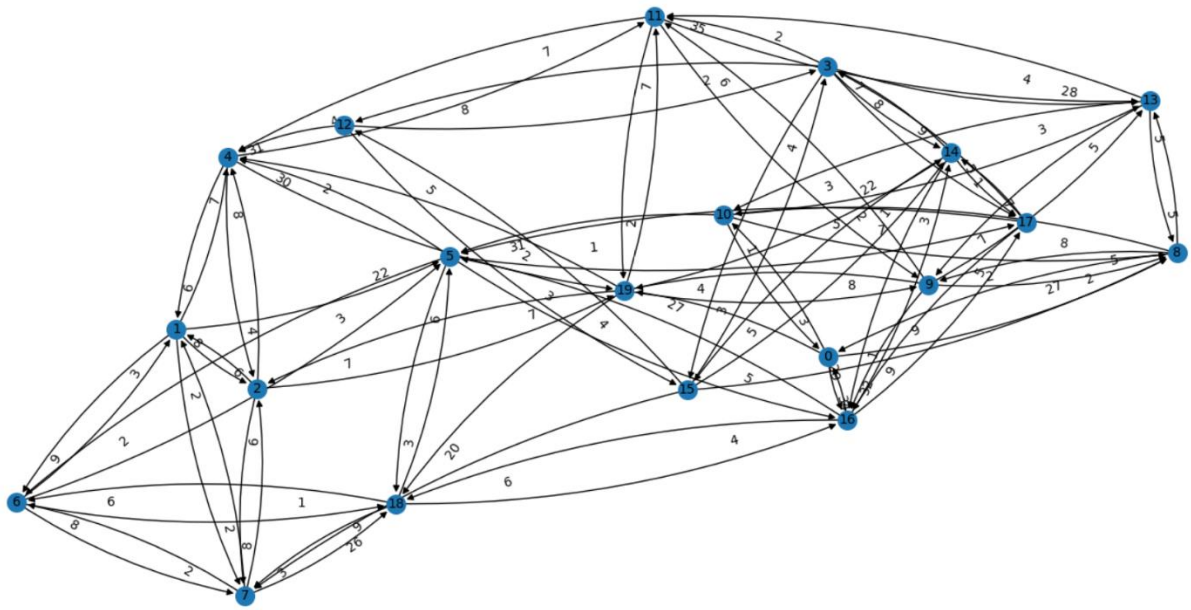
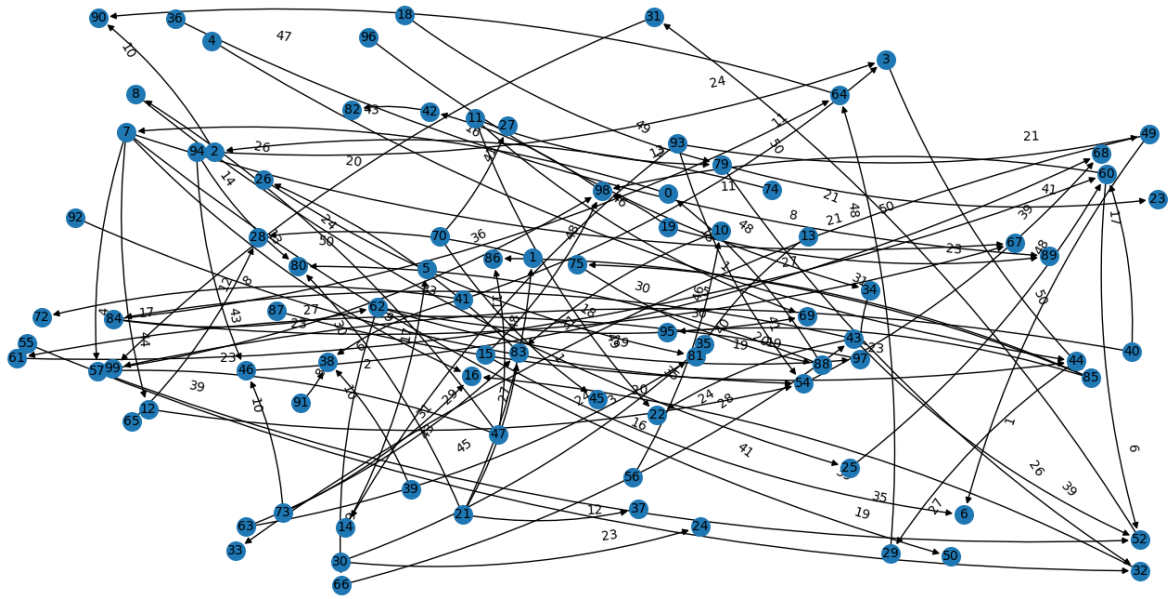
graph10.graphml



graph20.graphml



graph20.graphml (same graph different layout)



random_v10_e10_w50.graphml